

Nutrition vs Cost

What This Does

This product integrates USDA nutrition data with USDA Food-at-Home Monthly Area Prices to show how foods compare in nutrition per dollar.

Today, nutrition and price data exist separately; merging them makes cost-nutrition insights accessible and interoperable.

Users can search foods, view calories/protein per dollar, and compare categories using standardized USDA sources.

Wireframe

- Static Data:
 - Nutrition per 100g (USDA)
 - Monthly price per 100g (USDA FAH)
 - Precomputed metrics (calories/protein)
- Query:
 - Food name or category
 - Optional filters (region, month, nutrient type)
- Display:
 - “Here’s the nutrition profile and cost-efficiency for whole milk this month. Historically, here’s how its price and nutrient value compare to similar foods.”

Concepts

- Food Item
 - Nutrition (kcal, protein, other nutrients per 100g)
 - Price (per lb/unit » normalized to per 100g)
 - Category (controlled vocab aligned with USDA FAH)
 - Computed Metrics (calories/protein, cost/100g)

Structure

Backend: Python (Flask) (Or Node.js?) for data integration and metric computation

Frontend: HTML/CSS/JS or React for search + visualizations

Data: USDA Foundation Foods JSON + USDA FAH monthly price CSV

Architecture: Preprocessed tables (nutrition, prices, mappings, metrics), static charts, and lightweight API endpoints